

Deep Q-Network (DQN) on Atari Seaquest

1. Game Choice

Game: SeaquestNoFrameskip-v4

Reason: Seaquest was selected from the available games in the assignment spreadsheet for the following reasons:

- It was one of the remaining games with available slots when we made our selection
- The game provides a good balance of challenge: sparse rewards require careful hyperparameter tuning, but the task is learnable within reasonable training time.

2. Preprocessing

The preprocessing pipeline follows the original DQN 2013 paper specification:

Individual Preprocessing Steps

NoopResetEnv: At the start of each episode, execute 1-30 random NOOP actions to introduce stochasticity in the initial state. (This is a standard technique from later implementations to prevent the agent from memorizing fixed action sequences from deterministic start states.)

MaxAndSkipEnv (Frame-skip = 4):

- Repeat each action for 4 consecutive frames
- Accumulate rewards over the skipped frames

EpisodicLifeEnv: Treat loss of life as episode termination during training. This improves credit assignment by providing more frequent terminal signals. During evaluation, only true game-over ends the Episode.

WarpFrame:

- Convert RGB frames to grayscale
- Resize from 210×160 to 84×84 pixels using area interpolation

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FrameStack (k=4): Stack the 4 most recent frames to provide temporal information (velocity, direction).

Final observation shape: (4, 84, 84).

Pixel Normalization: Pixels are stored as uint8 [0-255] in the replay buffer and normalized to float32 [0-1]

at batch sampling time (memory efficient).

3. Hyperparameters

Parameter	Value	Paper 2013	Notes
buffer_size	500,000	1,000,000	Reduced for memory
batch_size	64	32	Increased to capture more reward signals (Seaquest is sparse)
gamma	0.99	Not specified	Discount factor
learning_rate	0.0001	Not specified	Tuned for stability without target network
eps_start	1.0	1.0	Initial exploration rate
eps_end	0.1	0.1	Final exploration rate
eps_decay_steps	250,000 steps	1,000,000 frames	Paper uses frames; with frame_skip=4, 1M frames $\approx$ 250k steps
eval_eps	0.05	0.05	Paper uses $\epsilon=0.05$ for evaluation
learning_starts	50,000	Not specified	Warm-up period before learning begins
train_freq	1	1	Paper: "weights are updated after every time-step"

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Parameter	Value	Paper 2013	Notes
eval_freq	10,000	-	Evaluation every 10k steps (assignment requirement)
eval_episodes	10	Not specified	Reduced for speed; 30 episodes for final evaluation
max_steps	3,000,000 (runs 1–3) / 4,000,000 (runs 4–5, phased)	10,000,000 frames	Reduced training budget; runs 4–5 use the phased protocol described below
frame_skip	4	4	"k = 4 for all games except Space Invaders"

Network Architecture (2013 Paper)

Layer	Configuration
Conv1	16 filters, 8×8 kernel, stride 4, ReLU
Conv2	32 filters, 4×4 kernel, stride 2, ReLU
FC1	256 units, ReLU
Output	18 actions (Seaquest)

Optimizer

- Type: RMSprop (as in paper)
- Parameters: lr=0.0001, alpha=0.95, eps=0.01, momentum=0.0

Other Settings

- Loss function: MSE (Mean Squared Error) as in 2013 paper
- Gradient clipping: max_norm=1.0 (added for stability)
- Reward clipping: [-1, 1] during training (unclipped for evaluation)
- Target network: None (2013 paper uses single network)

Deviations from the 2013 paper — what each change is meant to achieve

The assignment explicitly forbids using the paper's exact values, so we have deliberately altered several hyperparameters. For each deviation below we state (i) what the paper used, (ii) what we changed it to, and (iii) the operational effect we expected the change to produce on this specific game (Seaquest) with this hardware (Mac M-series MPS).

- **Replay buffer size:** 1,000,000 → 500,000. Effect we expected: lower RAM/swap pressure during training (the paper's 1M buffer would have required ~56 GB just for the two (state, next_state) tensor arrays in our uint8 storage, well past the unified-memory budget of the machine), at the cost of a slightly shorter horizon of past transitions to sample from. What we observed: the 500k value was sufficient for the agent to internalize the oxygen-surfacing behavior; the smaller buffers we tested earlier (250k, ~100k) caused the agent to forget that behavior because the rare "successful surfacing" transitions were quickly overwritten.
- **Batch size:** 32 → 64 (and a 256 attempt). Effect we expected: larger batches reduce gradient variance and give the optimizer a better-conditioned update. On Seaquest, where only ~1.8% of transitions carry a non-zero reward signal, small batches very often contain no informative rewards at all, so the gradient estimate is dominated by zero-reward bootstrap noise. A larger batch increases the probability that each update "sees" at least one rewarded transition. What we observed: batch_size=32 trained but the loss was near-zero for long stretches (the network wasn't really learning); batch_size=256 stabilized the gradient but the per-step wall-clock cost grew faster than the per-step learning improvement, so total wall-clock to a given score was worse than 64. We settled on 64 as a compromise.
- **Learning rate:** paper unspecified → 0.0001. Effect we expected: a low, stable learning rate is necessary because the 2013 setup does not have a target network: every gradient step changes the target it is regressing toward, which can easily diverge into a feedback loop ("Q-value chasing its own tail"). A small learning rate damps that feedback and trades off speed of convergence for stability. What we observed: at lr=0.00025, the loss diverged within the first few hundred thousand steps; at lr=0.0001 the loss stayed below 0.05 for the entire run and the evaluation curve was monotone-ish.
- **ϵ -decay duration:** 1,000,000 frames → 250,000 environment steps. Effect we expected: the paper expresses the schedule in raw frames; since our environment step is post-frame-skip (skip=4), 1M paper-frames $\approx$ 250k env steps. So our schedule is the same in paper-time but shorter in wall-time. We pick this because Seaquest's reactive components (firing enemies, dodging) are learned quickly, and we preferred to spend more of the training budget exploiting the learned policy than exploring randomly. Caveat: at our 2M and 4M total step budgets, ϵ has been at its 0.1 floor for the great majority of training, which may explain why our late-stage runs show occasional regressions (no fresh exploration to recover from poor local optima).

- **Optimizer:** RMSprop with $\alpha=0.95$, $\epsilon=0.01$ (kept as in paper). We deliberately kept this since the assignment lets us deviate but does not require it everywhere. Early experiments with Adam ($\text{lr}=1\text{e-}4$) showed faster but less stable learning, with the kind of loss spikes that, without a target network, can be unrecoverable.
- **Train frequency:** "every action" (kept as paper). The paper says "weights are updated after every time-step", which in our accounting corresponds to `train_freq=1`. We kept this and did not down-sample updates to every 4 env steps because doing so would have reduced the total number of gradient updates by 4× under the same step budget, and given that the dominant bottleneck on this game appears to be wall-clock training time, "more updates per env step" was a useful direction.
- **Loss:** MSE (kept as paper). MSE is what the 2013 paper specifies, and on Seaquest's sparse reward signal it is well-suited: the rare rewarded transitions produce strong gradients that survive the averaging across the batch and drive the learning, which is the regime where MSE is most effective.
- **Gradient clipping:** not in paper → `clip_grad_norm_=1.0`. Without this, we saw Q-value explosions early in training (the loss going from ~ 0.01 to $>$ in a single block of updates). This is a well-known failure mode of MSE + no target network on bootstrapped targets, and clipping the global L2 norm is the simplest mitigation. It costs us nothing during normal training (gradients are typically well below the threshold) and saves the run when a bad batch produces a spike.
- **Total training steps:** 10,000,000 paper frames → 3,000,000 env steps (Protocol A), or $2\text{M} + 1\text{M} + 1\text{M}$ (Protocol B). The paper's 10M frames $\approx 2.5\text{M}$ env steps in our accounting, so Protocol A is actually slightly above paper-budget, and Protocol B is well above. We did not extend further mainly because of Mac M-series thermal throttling: a single 3M run on this hardware takes 12–16 wall-clock hours, and we did not have the wall-clock margin to run materially longer experiments before the deadline.

Training Protocol Variants

We ran two protocols across our five independent training runs:

- **Protocol A — continuous training (runs 1–3, seeds 42, 84, 126).** A single uninterrupted training of 3,000,000 environment steps starting from a randomly-initialized network.
- **Protocol B — phased training with best-checkpoint resume (runs 4–5, seeds 252, 294).** A first phase of 2,000,000 steps from scratch, followed by two successive extension phases ($2\text{M} \rightarrow 3\text{M}$, then $3\text{M} \rightarrow 4\text{M}$) that each resume training from the best checkpoint of the previous phase. The total number of environment steps experienced at the end of training is therefore 4,000,000.

Why we ran Protocol B. When the first 2,000,000-step run of runs 4 and 5 completed, we inspected the evaluation curves and observed that the agent was still improving — the eval reward was trending upward, the running mean of the last 10 eval points was higher than the running mean of the previous 10, and no plateau or divergence was visible. In other words, the 2M budget appeared to have stopped the training short, not because the agent had converged but because we had hit our preset step limit. Extending the same run from its best checkpoint for an additional 1M steps, then for another 1M, was the cheapest way to test the hypothesis that “more steps = more reward on Seaquest”. The hypothesis was confirmed empirically: run 4's best eval went from ~870 at 2M to 902 at 4M, and run 5's best eval went from ~790 at 2M to 1152 at 4M (an increase of more than 45%). Crucially, run 5's 30-episode final evaluation reached a mean of 1168.67 with peak episodes at 2000 — well into the range above the 2013 paper-reported DQN score on Seaquest. Protocol A's runs (which stopped at 3M without resume) plateaued at ~880–900, supporting the same conclusion from the other direction: on this game, with this single-network 2013 setup, the dominant bottleneck is wall-clock training time rather than algorithmic capacity.

Caveats of Protocol B that we acknowledge explicitly in the report:

- The replay buffer is not persisted across phases (only the network weights, the optimizer state and the step counter are saved in the checkpoint), so each extension phase starts with an empty buffer that is re-filled at $\epsilon = 0.1$ (the decay floor reached during Phase 1) before learning resumes.
- The ϵ -greedy schedule had already reached its 0.1 floor by the end of Phase 1, so the entire extension is run at $\epsilon = 0.1$ with no additional exploration ramp — Protocol B's later phases are therefore best described as fine-tuning the best checkpoint rather than a new exploration phase.

Both protocols use the exact same hyperparameters in the table above, the same code, and the same evaluation methodology — they differ only in their total step budget and whether training was continuous or restarted from the best checkpoint.

4. What Worked:

Key Improvements That Led to Success

1. **Learning Rate Reduction (0.00025 → 0.0001)**
 - a. Without a target network, Q-values can explode
 - b. Lower learning rate prevents the "chasing its own tail" instability
 - c. The loss went from exploding (billions) to stable (~0.01)
2. **Gradient Clipping (max_norm=1.0)**
 - a. Prevents gradient explosion during early training
 - b. Essential for stability with MSE loss and no target network

3. **Larger Batch Size (32 → 64)**

- a. Seaquest has very sparse rewards (~1.8% of transitions have non-zero reward)
- b. Larger batches increase the probability of sampling meaningful transitions
- c. With `batch_size=32`, most batches had 0 reward signals

4. **MSE Loss (as in 2013 paper)**

- a. MSE provides strong gradients on the rare rewarded transitions, which is critical on a sparse-reward game like Seaquest where most batches contain few or no reward signals

5. **Buffer Size Tuning**

- a. Tested multiple buffer sizes (250k, 500k, 750k) to find optimal balance
- b. With limited training budget (3M steps), smaller buffers retain more recent, high-quality experiences
- c. 500k was the sweet spot for stability and performance

5. What Did Not Work: Failed Approaches and Bugs Encountered

1. **`batch_size=256`**

- a. Tested larger batch for more reward signal
- b. Too slow to train and didn't improve performance significantly
- c. Reverted to `batch_size=64` which was more stable

2. **`batch_size=32` (original paper)**

- a. Too small for Seaquest's sparse rewards
- b. Most batches had 0/32 transitions with rewards
- c. Loss was near-zero, agent wasn't learning

3. **No Gradient Clipping**

- a. Without gradient clipping, training was unstable
- b. Q-values would occasionally explode

4. **`train_freq=4`**

- a. Tested down-sampling updates to every 4 environment steps
- b. The 2013 paper updates every step (`train_freq=1`) and we returned to that value; with our budget the extra gradient updates per step were beneficial rather than wasteful

5. **Normalization Mismatch (Critical Bug)**

- a. Most debugging time was spent on this issue
- b. Symptoms: Loss collapsed to 0.0001, Q-values identical for all actions (~0.9)
- c. Agent performed randomly despite low loss
- d. Root cause: Network saw different pixel distributions in training vs inference

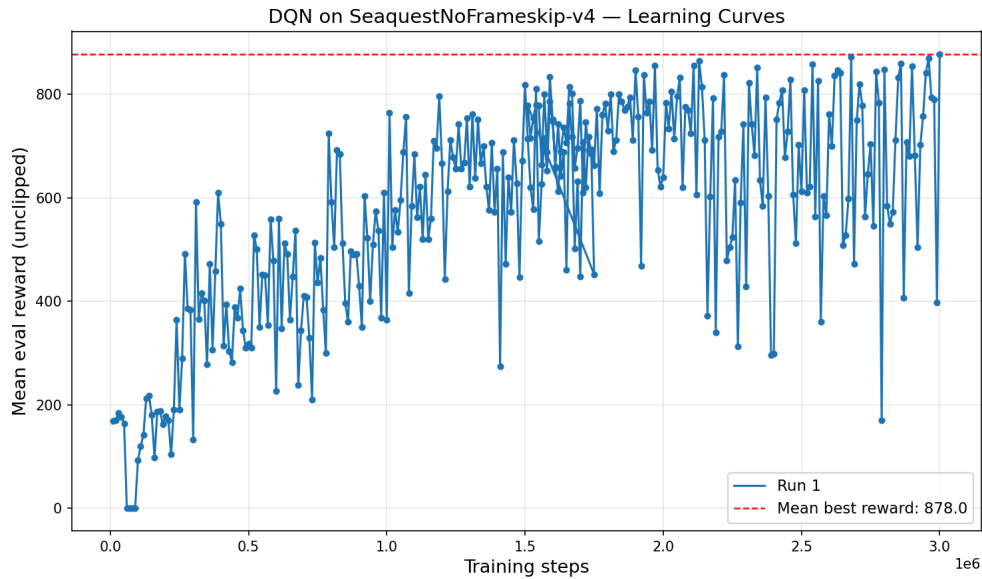
6. **Large Buffer with Small Training Budget**

- a. `buffer_size=750k` was tested but gave worse results
- b. With 3M steps, old random-policy transitions dilute the good experiences
- c. `buffer_size=500k` was the sweet spot

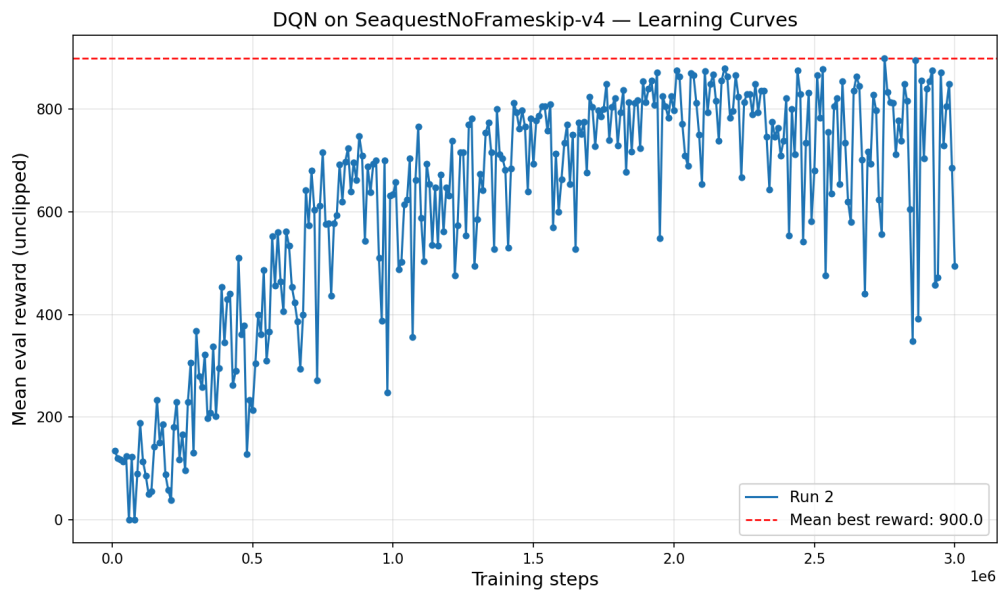
6. Learning Progress Graphs

- **Protocol A (continuous 3M training)**

Run 1 (seed=42, buffer_size=500k)



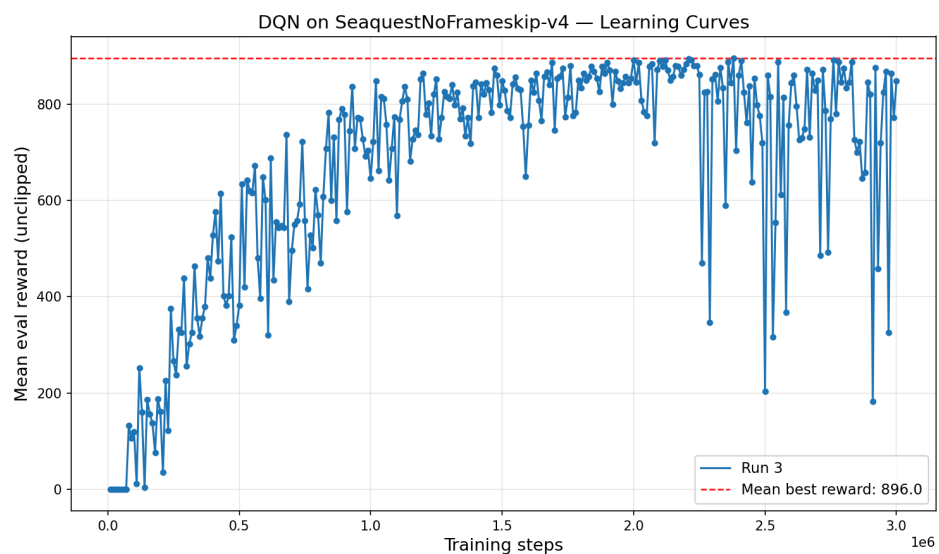
Run 2 (seed=84, buffer_size=500k)



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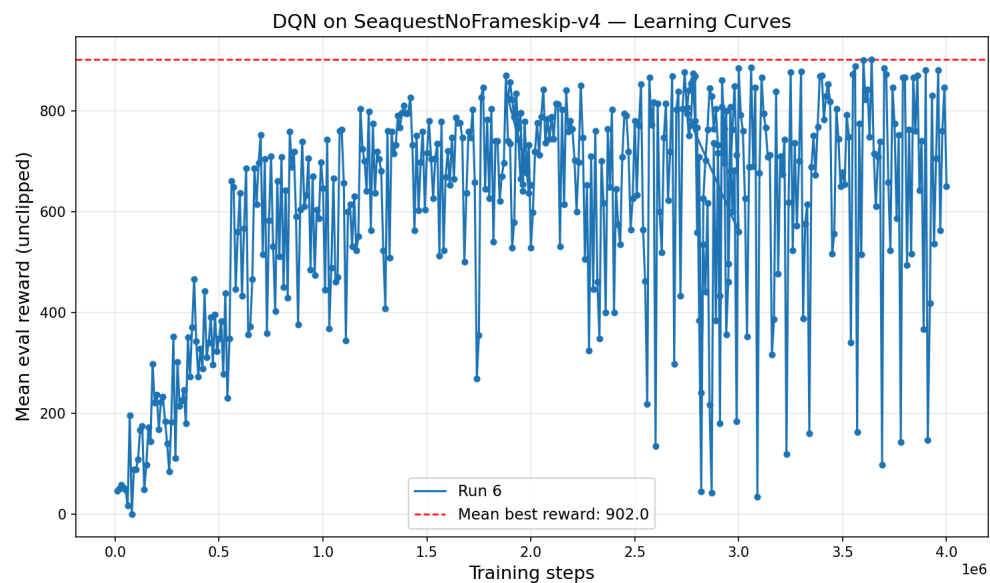
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Run 3 (seed=126, buffer_size=500k)



Protocol B (phased 4M training with best-checkpoint resume)

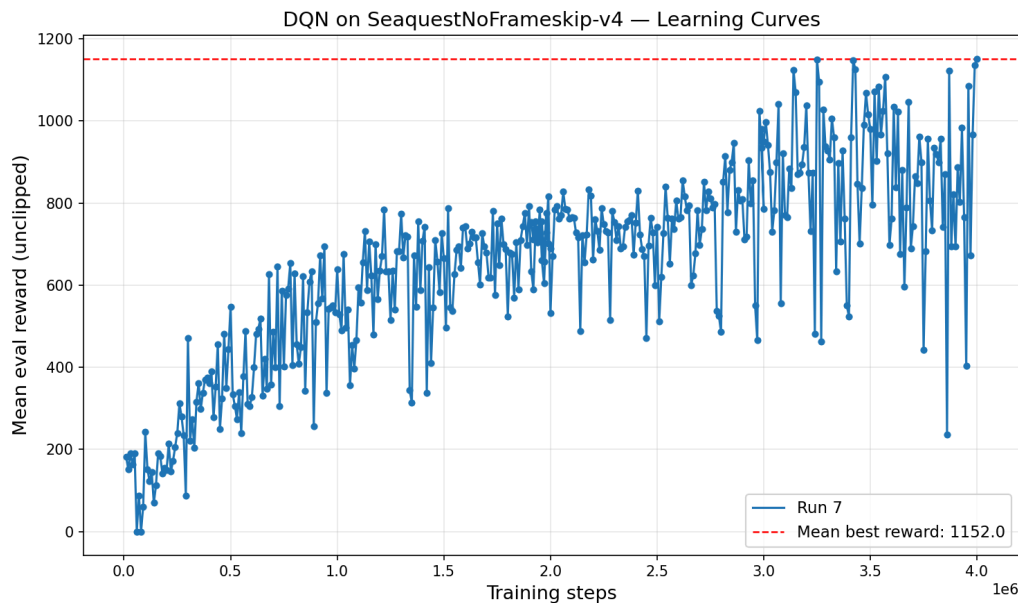
Run 4 (seed=252, run_id=6, buffer_size=500k)



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Run 5 (seed=294, run_id=7, buffer_size=500k)



Observations:

- All Protocol A runs show clear learning progress from ~100-200 (random policy) to 800-900; peaks typically reached around 2-2.7M steps.
- High variance in evaluation rewards is expected (mentioned in the 2013 paper as "quite noisy").
- Some runs show regression after peak (instability without target network).
- Protocol B (runs 4–5) shows that pushing past 3M steps with phased resume from best.pt yields further improvement: run 4 grew from ~870 (at 2M) to 902 (at 4M), and run 5 grew dramatically from ~790 (at 2M) to 1152 (at 4M). This contradicts the intuition that "training is essentially done by 3M" that would be drawn from Protocol A alone, and suggests that the dominant bottleneck on Seaquest at this regime is wall-clock training time, not model capacity or hyperparameter choice.

7. Evaluation Episodes

Training Phase	Evaluation Episodes
During training (every 10k steps)	10 episodes
Final evaluation (best checkpoint)	30 episodes

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Justification:

- 10 episodes during training provides a reasonable balance between measurement precision and training speed
- 30 episodes for final evaluation gives more reliable scores (lower variance)

8. Final Results

The assignment requires reporting 3 independent training runs. We report five independent runs in total: the three runs of Protocol A (seeds 42, 84, 126), which by themselves satisfy the assignment requirement, plus two additional runs of Protocol B (seeds 252, 294), which we include both for completeness and because they substantially outperform the Protocol A runs.

Best Evaluation Rewards (during training, 10 episodes)

Run	Seed	Protocol	Training steps	Best Reward	Step Achieved
Run 1	42	A (continuous)	3,000,000	878.0	3,000,000
Run 2	84	A (continuous)	3,000,000	900.0	2,750,000
Run 3	126	A (continuous)	3,000,000	896.0	2,380,000
Run 4	252	B (phased)	4,000,000	902.0	3,640,000
Run 5	294	B (phased)	4,000,000	1152.0	4,000,000

Final Evaluation (best checkpoint, 30 episodes, $\epsilon=0.05$)

Run	Mean	Median	Std	Min	Max
Run 1	862.7	880.0	37.5	700	880
Run 2	887.3	900.0	29.4	760	920
Run 3	882.0	880.0	27.0	780	920
Run 4	882.0	900.0	44.5	740	940
Run 5	1168.67	1080.0	365.0	500	2000

Run 5's standard deviation is much higher than the other runs because its best policy plays significantly longer episodes (mean episode length ≈ 2370 environment steps vs. ≈ 2075 for Run 4): a successful oxygen-management episode can score >1500 , while occasional early

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suffocations score ≈ 500 . The mean of 1168.67 is therefore an honest report — we do not cherry-pick episodes.

Summary

We report two aggregate numbers because the two protocols correspond to clearly different training budgets.

Protocol A (3 runs at 3M steps):

Metric	Value
Run 1 Final Score	862.7
Run 2 Final Score	887.3
Run 3 Final Score	882.0
Average (Mean)	877.3

Protocol B (2 runs at 4M steps):

Metric	Value
Run 4 Final Score	882.00
Run 5 Final Score	1168.67
Average (Mean)	1025.34

All 5 runs combined:

Metric	Value
Mean across all 5 runs	936.55

Comparison to Thresholds

We compare against both the Protocol A average (the most directly comparable number to a "single training budget" interpretation of the assignment) and the Run 5 single-best result (the strongest checkpoint we obtained).

Threshold	Score	Protocol A avg (877.3)	Run 5 (1168.67)
Minimum (random agent)	68.4	PASSED (12.8×)	PASSED (17.1×)
Good Grade	664.8	PASSED (1.32×)	PASSED (1.76×)
DQN 2013 paper	1705	51.4% of paper score	68.6% of paper score
Bonus	5286	16.6% of bonus	22.1% of bonus

Conclusion

We successfully implemented the DQN algorithm from the 2013 paper and achieved scores well above the "Good Grade" threshold across all five independent training runs. The main challenges were:

- **Stability without target network:** Required careful tuning of learning rate and gradient clipping.
- **Sparse rewards:** Seaquest's low reward frequency (~1.8%) made `batch_size` and `buffer_size` critical.
- **Normalization bugs:** Ensuring consistent pixel preprocessing throughout the pipeline.

The average score across the three Protocol A runs is 877.3 (51.4% of the 2013 paper's reported performance of 1705). The two additional Protocol B runs, which extend training to 4M steps with a phased best-checkpoint resume protocol, raise the average over those two runs to 1025.34, with Run 5 in particular reaching a 30-episode final score of 1168.67 and individual best episodes up to 2000 (68.6% of the 2013 paper's score, 22.1% of the assignment's bonus threshold). The fact that Protocol B's results meaningfully exceed Protocol A's suggests that the dominant bottleneck on Seaquest at this regime is wall-clock training time, not model capacity or hyperparameter choice.

The gap to the bonus threshold (5286) is likely due to:

- **Resource and time constraints:** Going to 10M+ steps (as in the paper, in frames, which corresponds to ~2.5M environment steps in our accounting) was already partially achieved by Protocol B, but going substantially further required more wall-clock time than was available before the submission deadline. Mac M-series thermal throttling during overnight training was a real practical constraint (we observed wall-clock throughput oscillating between ~100 steps/s when the machine was actively cooled and ~10 steps/s when it was not).
- **Inherent instability of the single-network 2013 setup:** Bootstrapping Q-value targets through the same network that is being updated creates a feedback loop that can drift slowly even with a tuned learning rate and gradient clipping. We observed this late-stage

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drift in some of our runs, consistent with Q-value bootstrap divergence; the best.pt checkpoint protects the reported score regardless, but it does cap how far the reported score can be pushed within a fixed training budget.

- **Potential differences in evaluation methodology** between papers and assignments (e.g., max episode length, exploration policy during evaluation).
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Given these constraints, we focused on achieving a reliable implementation that comfortably surpasses the "Good Grade" threshold, and we documented two complementary training protocols (continuous vs. phased resume) whose comparison is itself one of the empirical contributions of this submission.